Unit Plan: Forensic Chemistry

FOR THE TEACHER

Created by AACT

Summary

The AACT high school classroom resource library and multimedia collection has everything you need to put together a unit plan for your classroom: lessons, activities, labs, projects, videos, simulations, and animations. We have constructed this unit plan using AACT resources designed to review key chemistry concepts with your students through a forensic chemistry lens.

Grade Level

High School

NGSS Alignment

The teaching resources used in this unit plan will help prepare your students to meet performance expectations in the following standards:

- **HS-PS1-3**: Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.
- **HS-PS1-8**: Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay.
- **HS-PS2-6**: Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.
- **HS-PS3-2**: Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects).
- **HS-PS4-1**: Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media.
- **HS-PS4-3**: Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model, and that for some situations one model is more useful than the other.
- **HS-PS4-4**: Evaluate the validity and reliability of claims in published materials of the effects that different frequencies of electromagnetic radiation have when absorbed by matter.
- **HS-ETS1-2**: Design a solution to a complex real-world problem by breaking it down into small, more manageable problems that can be solved through engineering.
- **HS-ETS1-3**: Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.

- **Science and Engineering Practices:**
  - Developing and Using Models
  - Analyzing and Interpreting Data
  - Engaging in Argument from Evidence
  - Constructing Explanations and Designing Solutions
  - Using Mathematics and Computational Thinking
  - Obtaining, Evaluating, and Communicating Information
  - Planning and carrying out Investigations.

Objectives

By the end of this unit plan, students should be able to:

- Read and interpret a mass spectrum.
- Identify an unknown piece of glass using physical properties of matter.
- Compare patterns in mock samples of DNA.
- Design and conduct an interview.

- Make and support a claim about a crime scene based on analysis of evidence.
- Understand how scientists use half-lives and carbon dating to try to determine the age of fossils and rocks.
- Interpret results of a flame test.
- Explain how movement of electrons results in the absorption of light.
- Separate component pigments found in a marker.
- Identify the component pigments of a black marker.
- Calculate the Rf value of the component pigments.
- Explain how chromatography can work to separate the components of a solution.
- Explain what information can be determined from GC-MS.
- Identify an unknown gas based on experimental evidence.
- Identify an unknown metal based on specific heat capacity values determined.
- Create and interpret a chromatogram, by separating the mixture of lipstick, to identify unknowns.
- Calculate the molarity of acid in a water source and analyze that data to determine where a drowning occurred.
- Calculate the molarity of an unknown substance.
- Demonstrate how over-the-counter drugs react differently to chemical tests and use that information to identify an unknown drug.
- Recognize relationships between temperature, volume and pressure of a gas.

Chemistry Topics

This unit plan supports students' understanding of:

- Physical Properties of Matter
- Density
- Nuclear Chemistry
- Radioactive Isotopes
- Energy
- Solutions
- Concentration
- Spectroscopy
- Specific Heat Capacity
- Molecular Structure
- Intermolecular Forces
- Polarity
- Separating Mixtures
- Chemical Reactions
- Identifying an Unknown
- Acids and Bases
- Titrations
- Gas Laws

Time

Teacher Preparation: See individual resources.
Lesson: See individual resources.

Materials

- Refer to the materials list given with each individual activity.

Safety

- Refer to the safety instructions given with each individual activity.

Teacher Notes

- Teachers should note that some of the resources in this unit plan were written for AP Chemistry. However, most can easily be adapted for an on-level or honors class.
- The resources selected for this unit plan are listed in the suggested order that they should be completed.
- Teacher notes, student handouts, and additional materials such as answer keys and expected results can be accessed through the link on the individual webpage for each activity.
- Please note that most of these resources are AACT member benefits.
- This unit plan was designed to review first year chemistry content through the lens of forensic science. You will notice some core content areas are missing, as they may not lend themselves to this approach as easily. Alternatively, teachers can choose to select resources from this unit plan to implement during their chemistry curriculum, making forensic science connections throughout the curriculum, rather than as a year-end review.
- Several *ChemMatters* Magazine "Mystery Matters" articles are included in this unit plan. While they are from a column that appears in only older issues of *ChemMatters*, the content is still relevant today.
    - Each Mystery Matters article shares the story of a crime investigation and how chemistry played a role in providing forensic evidence and solving the case.
    - Use the corresponding Teacher's Guides for all *ChemMatters* issues to accompany these activities with anticipation questions, guided reading comprehension questions, graphic organizers and more if time allows. All *ChemMatters* Teacher Guides are available for download from the issue pages in the AACT *ChemMatters* archive. Alternatively, Teachers Guides for the most recent issues can also be found on the ACS *ChemMatters* webpage.
- Before reading the outline of our Forensic Chemistry unit plan below, it can be helpful to have some resources on how other teachers have applied forensic science themes in their chemistry classrooms. If you have time, the following resources give some great ideas to keep in mind as you plan:
    - AACT *Chemistry Solutions* article: Recasting Chemistry Labs with Forensic Themes
    - AACT Webinar: Incorporating Forensic Science into the Chemistry Curriculum

Classroom Resources

Introduction to Forensic Chemistry:

- Begin the unit with a short introduction to forensic chemistry. Before having students experience a forensics-based lab, they should watch a video that explains the role of a forensic scientist at a crime scene. The video, TV Forensics: What Do CSIs Actually Do?, from the ACS Reactions video series is a good option.

- Next, students review fundamental chemistry topics with a focus on evidence. In the lesson, the Shattered Glass Mystery students take on the role of a Forensic Scientist to help solve a hit and run investigation. They will learn how physical properties of matter, such as density and refractive index, can be used to help identify evidence samples such as glass. Additionally, the October 2006 *ChemMatters* article, "Glass: More than Meets the Eye" explains concepts from the Shattered Glass Mystery lesson in depth if you would like to incorporate more literacy.

- Another option is to modify the lab, Strawberry DNA & CSI, originally created for younger students, to instead use in the high school classroom. Students extract DNA from strawberries and analyze evidence to figure out who perpetrated a petty crime. They will compare sets of DNA, test pH, and design and conduct interviews to crack the case.

- To increase student curiosity, teachers can incorporate additional short chemistry videos themed in Forensic Chemistry. Two recommended videos from the ACS Reactions series are:
    - Exposing Fake Honey using Carbon-13 (~12 minutes)
    - Can Science Beat Counterfeit Detection Pens? (~10 minutes)

Elements and the Periodic Table:

- Use Sam Kean's Arsenic video to engage students with a mystery. This would be great as a bellringer activity. Teachers can first present a fictitious scenario that has either a deceased victim with an unknown cause of death or a mysterious substance that was found that has led to someone's death. Arsenic's atomic number or atomic mass could be included as a "strange" number that was found written in a notebook near the crime scene. This allows students to think critically and use problem solving skills, possibly leading them to the periodic table to find an explanation. Show the arsenic video (~4 minutes) after students have offered solutions to the mystery.

Nuclear Chemistry:

- Teachers can review isotopes and nuclear stability in one dedicated class period. It's recommended to choose between these two lab activities:
    - Radioactive Dating: The Demise of Frosty has students investigate carbon dating to extrapolate data and ultimately determine if Frosty the Snowman started melting in class and which culprit was responsible for poor Frosty's demise.
    - Using Stable Isotopes to Determine Material Origin requires students to review isotopes and apply the concepts of stability and relative abundance in order to determine the recent travels of a person of interest in a criminal investigation
- Follow up the activity with a short review on isotope symbols, what decay is, half-lives, and types of decay.
- Finally use the video, Real-life CSI: Age dating fingerprints from ACS Headline Science or the *ChemMatters* Mystery Matters article, "Forensics Finding the Chemical Clues" to complete review of this topic.

Electrons and Light:

- Use the lab, Determining the Time of Death to engage students with a twist on the traditional flame test lab. Students will examine a sample of "vitreous humor" and identify which element from the sample is used to determine the time of death. Then they will engineer a simple spectrophotometer to quantify that element. Evaluating a fake sample of vitreous humor in their spectrophotometer will help them determine the time of death for a hypothetical cadaver. Since this also introduces spectrophotometry, it serves as a great introduction to how the various types of spectroscopies are used in forensics cases.
- If time allows, follow the lab the short 3-minute video, How Does Fluorescence Work?, from ACS Reactions. Alternatively, students can read a recommended *ChemMatters* magazine article "Authentic or Not? Chemistry Solves the Mystery" or The Mystery Matters, "The Cattle Killer" to further investigate this topic.

Bonding and Intermolecular Forces:

- A lab focused on polarity and chromatography can be a simple introduction into more complex types of chromatography that forensic scientists use. Have students demonstrate their knowledge of the effects of polarity and solubility using the lab, Using Paper Chromatography to Separate the Pigments Found in Ink.
- Follow this with other types of chromatography, such as the Landmark Lesson, GC-MS (Gas Chromatography-Mass Spectrometry) or by introducing high performance liquid chromatography. Emphasize that even though more expensive machinery is needed, the basic premise of using polarity and other properties of compounds to separate them remains the same.

- Related *ChemMatters* articles from the Mystery Matters series that could be included here:
    - Dog Gone: There was no weapon, no sign of a struggle, and no marks on the body. What caused the death?
    - The Forgery Murders: A true story of greed, forgery, deceit, murder…and chemistry.
    - Good Science Goes Bad: Did a mother really poison her baby with ethylene glycol?

Chemical Reactions and Stoichiometry:

- Use the Priestley labs to review chemical reactions with a forensic theme. Students identify an unknown gas produced from the reaction of hydrogen peroxide and manganese dioxide. Talk about the signs of a chemical change and what this lab taught them about predicting products. As a bonus, have them calculate the theoretical mass of the gas produced. To increase the forensics science investigation, replace the lab conclusion with a crime scene report that determines the identify of the mystery substance.
- If time allows, read the *ChemMatters* article, "The Forensics of Blood" as a class and record all the ways that chemical reactions are useful to forensic scientists. Additionally students can read the Mystery Matters article, "*Nightmare on White Street*" to learn more about this topic applied to a real crime.

Thermodynamics:

- Use the lab, The Search for a Hit and Run Suspect, to challenge students with solving a mystery using specific heat capacity. They will investigate three metals found at the crime scene and link their analysis of the metals to help determine if a suspect's vehicle was potentially involved in a hit and run incident.

Solutions:

- The lab, Chemicals, Chromatography, and Crime! tasks students with testing "evidence" that has been collected from a crime scene to determine if a victim was poisoned. They will perform a solubility and crystallization test on an unknown powder and attempt to identify the culprit by using paper chromatography to analyze the lipstick from the potential criminals.
- To further support the topic of solutions, specifically the importance of concentration, students can read about a story that examines the remains of a perished sailors from the doomed Franklin Expedition in the *ChemMatters*, Mystery Matters article, Buried in Ice.

Acids and Bases:

- Teachers can focus on titrations and molarity calculations using either the lab, Investigating a Suspicious Drowning with Titrations or Lethal Dose. In the Suspicious Drowning lab, students conduct a series of titrations on an evidence sample of water collected from the victim's lungs as well as on several water samples from local water sources (lakes, rivers, wells, etc.) to determine where the victim drowned. In the Lethal Dose lab, students perform several simple titrations to calculate the concentration of potentially "lethal" medicycloprophic acid and determine if the concentration makes the dosage lethal.
- An additional option is to use the lab, Over the Counter Drugs, to focus on pH and acid and base properties. Students conduct several chemical tests on over-the-counter medications and use their results to identify an unknown drug sample.

Electrochemistry:

- As a quick review of the electron transfer and types of reactions used in electrochemistry, allow students time to read the article, Blood Markers, from the Mystery Matters column of *ChemMatters* magazine. In this article, forensic scientists reconstruct a crime using chemistry and the evidence from a dead woman, a wounded man, a bloody knife, and a smudged pack of cigarettes.

Gases:

- Consider reviewing gas laws with reference to the Titan submersible implosion that occurred in 2023. Frame it as a mystery as to why the submarine never resurfaced. There are many news articles and videos that can be used to engage students with this high-profile case. CBS News: Search intensifies for missing Titan submersible is a good option, but it can be replaced with another video if needed.
- This will lead to students investigating the effects of pressure changes outside of the vessel versus the pressure inside the vessel. You can accompany this with the Crush the Can demo, though be aware of how this visualization tied in with the knowledge of what happened to occupants of the submarine may impact students.

Wrap-Up:

- End the unit with a lab that encompasses multiple techniques to solve a mystery. Consider The Culminating Unknown where students identify an unknown from a list of 12 possible compounds by designing a procedure and using evidence to prove their claim. Alternatively, the lesson, How Modern Instrumentation Revolutionized the Poison Game tasks students with creating their own Safety Data Sheet on a poison of choice, and learning about how mass spectroscopy has helped revolutionize the modern analysis of toxins. Teachers should note that these options require more than one class period, so be sure to consult the teacher's notes and required time before you incorporate them.

Additional Activities for consideration:

- Students interested in forensics will enjoy diving into these readings, puzzles, and activities.
    - Activity: Forensic Chemistry Crossword
    - *ChemMatters* magazine article: Four Cool Chemistry Jobs (includes a forensic scientist)
    - *ChemMatters* magazine article: How Reliable is Forensic Evidence?
    - *ChemMatters* magazine article: The Future of Forensics
    - The 2016 issue of *Celebrating Chemistry* magazine includes activities, articles, and labs, themed in forensic chemistry